

Software Requirements Specification (SRS)

Project: LearnLens AI

Tagline: "Show your AI what you are learning."

1. Introduction

LearnLens AI is an advanced, local-first personal learning environment and multimodal AI teacher. The system allows a human learner to explicitly grant observation access to an active Chrome browser tab, window, or desktop display. LearnLens AI continuously captures, transcribes, analyzes, and indexes educational content into a private, session-scoped knowledge base, providing an interactive, ChatGPT-like personalized tutoring experience.

2. Purpose

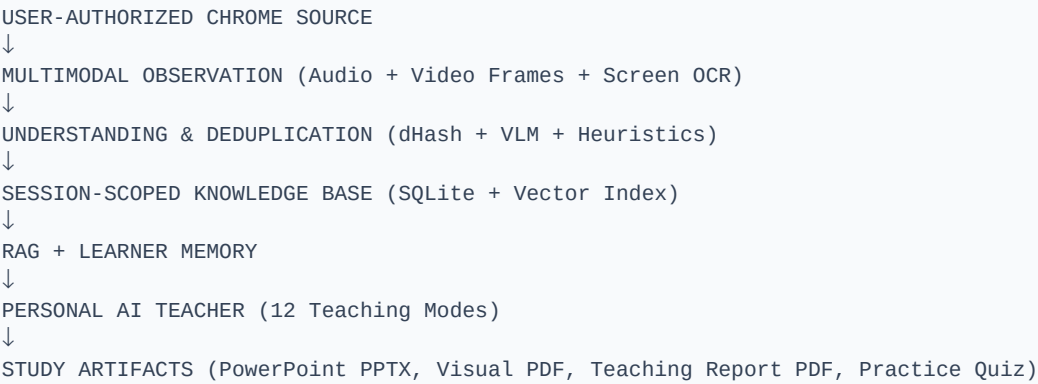
The primary objective of LearnLens AI is **accelerated human learning**. Traditional educational video consumption requires learners to watch hours of lectures, manually pause for note-taking, search across slides for diagrams, and struggle to locate answers to specific doubts. LearnLens AI transforms user-authorized learning sources into structured, searchable knowledge, automated slide presentations (PPT), comprehensive study guides (PDF), and grounded interactive AI tutoring.

3. Scope

LearnLens AI functions as a universal educational intelligence engine independent of specific platforms (such as Coursera, YouTube, Fortinet Training Institute, Forage, or Udemy). It operates locally on the user's workstation, utilizing local-first inference for audio transcription (faster-whisper), computer vision / OCR, session-scoped vector retrieval (RAG), and local language models via Ollama or built-in heuristic fallbacks.

4. Product Perspective

LearnLens AI sits alongside the user's browser environment:



5. Product Functions

- Universal Source Capture:** Native browser display media picker for Chrome tabs, windows, or screen.
- Local Audio Transcription:** Timestamped, segment-level speech-to-text with speaker identification.
- Smart Screenshot Intelligence:** Perceptual difference hashing (dHash) to filter redundant frames.
- Visual Classification & OCR:** Automated categorization into DIAGRAM, SLIDE, CODE, TABLE, or DEFINITION.
- Session-Scoped RAG:** Partitioned knowledge namespaces preventing cross-contamination across unrelated courses.

6. **AI Teacher Dialogue:** 12 pedagogical modes with exact timestamp citations and slide thumbnail evidence.
7. **Human Interrupt & Control:** Prominent Pause, Resume, Stop, and natural language command execution.
8. **Automated Presentation Generation:** Python-pptx generation with 5 professional layout styles.
9. **Dual PDF Publication:** Visual Study Pack (PDF A) and AI Teaching Report (PDF B).
10. **Quiz PDF Import & Practice Mode:** Formative assessment preparation without unauthorized test botting.
11. **Learner Memory Tracking:** Mastery profiling, doubt tracking, and knowledge gap remediation.
12. **Offline Video Processing:** Local ingestion of MP4, MKV, WebM, and MOV video files.

6. User Classes

- **Student / Self-Directed Learner:** Primary user studying online courses, technical documentation, or lectures.
- **Certification Candidate:** Learner preparing for high-stakes exams (e.g., Fortinet NSE, AWS, CompTIA).
- **Educator / Mentor:** User generating structured PPT decks and revision packs from raw instructional material.

7. Operating Environment

- **Operating Systems:** Windows 10/11, macOS 12+, Linux (Ubuntu 20.04+).
- **Browsers:** Google Chrome 110+, Chromium-based browsers (Edge, Brave).
- **Runtimes:** Python 3.10+, Node.js v18+.

8. Functional Requirements (FR)

| Identifier | Requirement Statement |

| :--- | :--- |

| **FR-01** | The system shall allow the user to initiate a new learning session with custom title and platform metadata. |

| **FR-02** | The system shall invoke the browser's native `getDisplayMedia` picker to obtain explicit user permission for tab/window capture. |

| **FR-03** | The system shall capture permitted video frames at periodic intervals and stream them to the local processing pipeline. |

| **FR-04** | The system shall capture audio from the authorized source and generate timestamped transcript segments locally. |

| **FR-05** | The system shall compute perceptual difference hashes (dHash) to automatically deduplicate stationary or redundant frames. |

| **FR-06** | The system shall perform Optical Character Recognition (OCR) on extracted visual frames. |

| **FR-07** | The system shall extract structured concepts containing name, definition, mechanisms, evidence timestamps, and exam importance. |

| **FR-08** | The system shall maintain session-scoped vector indices and inverted keyword tables for local RAG retrieval. |

| **FR-09** | The system shall provide an interactive ChatGPT-style conversational tutor interface. |

| **FR-10** | The system shall ground all tutor explanations in captured source timestamps and display clickable thumbnail evidence. |

| **FR-11** | The system shall allow the human user to pause or interrupt agent processing at any moment. |

| **FR-12** | The system shall allow the human user to resume paused processing without losing existing session state. |

| **FR-13** | The system shall generate structured PowerPoint (.pptx) presentations containing captured visuals and key insights. |

| **FR-14** | The system shall compile and export publication-ready PDF documents (Visual Study Pack & AI Teaching Report). |

| **FR-15** | The system shall import assessment quiz PDFs and extract questions, options, and explanations. |

- | **FR-16** | The system shall provide an interactive practice mode that records user answers and diagnoses weak concepts. |
- | **FR-17** | The system shall allow pinning of concepts, messages, and screenshots to the study dashboard. |
- | **FR-18** | The system shall support global keyword and concept search across all saved learning sessions. |
- | **FR-19** | The system shall allow users to delete any session, purging all associated database records, frames, and exports. |
- | **FR-20** | The system shall ingest and process local offline video files (MP4, MKV, WebM) without uploading data to external clouds. |
- | **FR-21** | The system shall provide 12 specialized pedagogical modes (Simple, Detailed, Exam Focus, Active Recall, Flashcards, etc.). |
- | **FR-22** | The system shall interpret natural language commands ("Pause", "Generate PPT", "Explain what I learned", "Make notes"). |
- | **FR-23** | The system shall provide real-time bidirectional WebSocket event synchronization for transcripts, frames, and agent states. |
- | **FR-24** | The system shall include an instant seedable educational demo session for evaluation without requiring external model servers. |

9. Non-Functional Requirements (NFR)

- **NFR-01: Privacy:** All captured video, audio, transcripts, and database tables must reside strictly on the local machine.
- **NFR-02: Performance:** Frame deduplication must complete in under 50 milliseconds per frame on standard quad-core CPUs.
- **NFR-03: Responsiveness:** RAG retrieval and initial tutor synthesis must begin within 1.5 seconds of user query submission.
- **NFR-04: Availability:** The application must function completely offline when using local heuristic synthesis or local Ollama instances.
- **NFR-05: Modularity:** Model providers (LLM, VLM, OCR, Transcription, Embedding) must adhere to abstract base class interfaces.

10. External Interfaces

- **Web UI:** Modern single-page application built with React, TypeScript, and Tailwind CSS.
- **REST API:** HTTP JSON API exposed via FastAPI at `http://127.0.0.1:8000/api`.
- **WebSocket:** Bidirectional JSON streaming endpoint at `ws://127.0.0.1:8000/ws/session/{session\_id}`.
- **Chrome Extension:** Manifest V3 extension featuring background service worker and responsive side panel.

11. Hardware Requirements

- **Minimum:** Dual-core x86_64 or Apple Silicon CPU, 8 GB RAM, 2 GB available disk space.
- **Recommended:** 8-core CPU, 16 GB RAM, NVIDIA RTX GPU with 6 GB+ VRAM for accelerated local VLM/Whisper models.

12. Software Requirements

- Python 3.10 or higher.
- Node.js v18.0 or higher.
- Modern Chromium-based browser (Chrome, Edge, Brave).

13. AI Requirements

- Pluggable LLM interface supporting local Ollama (`qwen2.5`, `llama3`) and zero-dependency heuristic synthesis fallback.

- Pluggable VLM interface supporting vision models (`qwen2.5-vl`, `llava`) and OpenCV contour complexity heuristics.
- Pluggable transcription interface supporting `faster-whisper` and browser Web Speech API streaming.

14. Browser Integration Requirements

- Must utilize standard `navigator.mediaDevices.getDisplayMedia` API.
- Must clearly prompt user with native browser picker.
- Must honor tab audio capture permissions.

15. Data Requirements

- Relational schema stored in SQLite (`data/learnlens.db`).
- File storage in partitioned session folders (`data/sessions//...`).
- Automated JSON serialization of array attributes.

16. Security Requirements

- Zero storage or transmission of user browser cookies, session tokens, passwords, or authentication credentials.
- Input validation on all API endpoints using Pydantic models.
- CORS restricted to localhost origins and Chrome extension protocol.

17. Privacy Requirements

- User must explicitly choose which tab or window is shared.
- Video stream immediately terminates when the user clicks "Stop Sharing" or the application's "Stop" button.
- One-click total purge of session data and disk files.

18. Performance Requirements

- Support continuous capture of educational lectures up to 4 hours in duration.
- Memory consumption must not exceed 1.2 GB during full multimodal processing.

19. Reliability Requirements

- Graceful degradation: if Ollama or GPU is unavailable, system must automatically switch to local heuristic tutoring without crashing.
- Database transactions must maintain ACID compliance.

20. Usability Requirements

- Clean, dark-mode ChatGPT-style UI with intuitive iconography.
- Clickable timestamp links that instantly jump to lecture slides.
- No developer jargon or technical debug logs exposed in the primary user interface.

21. Constraints

- System must run on local commodity hardware without mandatory cloud subscriptions.
- Browser security restrictions prevent silent capture without user interaction.

22. Assumptions

- The user has authenticated into their target educational website in Chrome prior to launching capture.
- The user's audio output device routes tab sound through the shared display stream.

23. Dependencies

- FastAPI, Uvicorn, SQLite3, python-pptx, ReportLab, PyMuPDF, OpenCV, Pillow, PyTesseract, React, Vite, Tailwind CSS.

24. Acceptance Criteria

- Full verification of the 40-step acceptance test: from tab capture, Whisper transcription, and slide deduplication to AI Teacher tutoring, PPT generation, PDF generation, and quiz practice gap diagnosis.